NATIONAL UNIVERSITY
BANGLADESH



First Year Syllabus

Department of Chemistry

Four-Year B. Sc. (Honors) Program

Effective from the Session: 2024 – 2025

Mission

To promote "Progress for Everyone" by fostering transformative education, conducting pioneering research, and encouraging meaningful collaboration with individuals, communities, and partners both in Bangladesh and around the world.

Vision

We expect that the hundredth anniversary of the National University, our commitment to "Progress for Everyone" will be evident in all areas of our operations and stakeholder engagements.

We leverage our core strength in education, research, and community engagement to address pressing global issues and foster sustainable development.

By 2042, National University aims to strengthen its positions as Bangladesh's top academic institution and rank among the world's leading universities. Our focus will be on expanding expertise building strategic partnerships, and increasing our global influence. Our services will be measured by the achievements of the individuals and communities we serve.

Name of the Program: B.Sc. (Honours) in Chemistry

Program Educational Objectives (PEOs):

After graduation, our students will be able to:

Objectives	Description	Domain
PEO 1	To provide knowledge in all the major branches of chemistry to understand the key concepts, principles and theories associated with life nature and society.	Fundamental
PEO 2	To provide the opportunity of conducting numerous scientific experiments with effective scientific communication skills.	Social
PEO 3	To equip the graduates with research-oriented skills including research proposal writing, research design, method validation and analyses, report writing and presentation of research outcome.	Thinking
PEO 4	To develop problem-solving, group work and communication skills so that they can operate with a high degree of autonomy.	Fundamental
PEO 5	To develop professionalism, resilience, and morality as well as to infuse an understanding of the culture, sustainability and human rights.	Personal

PEO = Program Educational Objective

Program Learning Outcomes (PLOs):

After successful completion of the Chemistry Graduate degree, the learner will be able to:

A. Fundamental Skills	
A1	formulate and explain the fundamental concepts in the core areas of the discipline (organic, inorganic, analytical, physical, industrial, pharmaceutical and/or biological chemistry).
A2	identify and locate current, reliable and accurate sources of scientific knowledge and apply knowledge and relevant skill to solve chemistry-related problems.
A3	employ critical thinking and scientific inquiry into the design, performance, interpretation and documentation of laboratory experiments at a level suitable to succeed at an entry-level position in chemical industry or a chemistry graduate program.
A4	work effectively with others in various situations, including a laboratory setting, classroom setting, or a team research project.
B. Social Skills	
B1	clearly communicate the results of the scientific work in oral, written, and electronic formats to both scientists and the public at large.
B2	convey ideas and information effectively to a range of audiences for a variety of purposes and contribute in a positive and collaborative manner to achieve common goals.
B3	participate in career planning, club activities, societies, co-curricular activities, internships, and different social as well as workplace settings and demonstrate the ability to incorporate entrepreneurial skills within the scope of different activities.
C. Thinking Skills	
C1	use critical thinking skills and scientific reasoning to evaluate arguments and address complex issues based on facts, experiments and observations.
C2	demonstrate the ability to think in an integrated manner and look at problems from different perspectives.
C3	design and carry out scientific experiments as well as accurately record and analyze the results of such experiments in both chemistry and allied fields of science and technology.
D. Personal Skills	
D1	engage in self-guided learning.
D2	appreciate the central role of chemistry in our society and use this as a basis for ethical behavior in issues facing chemists including an understanding of safe handling of chemicals in contemporary environmental and global issues.

D3	demonstrate discipline-specific computer skills, including use of software for word processing, spreadsheet calculations, database applications, graphing and curve fitting, data acquisition, and instrument control.

PLO = Program Learning Outcome

Mapping PLOs with PEOs:

Program Education Objectives (PEO)	**Program Outcomes (PLOs)**												
	Fundamental Domain				**Social** Domain			**Thinking** Domain			**Personal** Domain		
	A 1	A 2	A 3	A 4	B 1	B 2	B 3	C 1	C 2	C 3	D 1	D 2	D 3
PEO 1	•	•				•						•	
PEO 2	•	•				•		•	•	•		•	•
PEO 3		•	•					•	•	•	•		•
PEO 4		•	•	•	•	•	•		•	•		•	•
PEO5				•		•	•		•		•	•	

Mapping Course with PLOs: **FIRST YEAR**

FIRST YEAR CHEMISTRY COURSE CODE	**PLOs**												
	Fundamental Domain				**Social** Domain			**Thinking** Domain			**Personal** Domain		
	A 1	A 2	A 3	A 4	B 1	B 2	B 3	C 1	C 2	C 3	D 1	D 2	D 3
First Year													
212801	•	•				•						•	
212803	•	•				•		•	•	•		•	•
212805		•	•					•	•	•	•		•
212806	•	•	•	•	•	•	•		•	•		•	•
213709	•	•				•		•		•		•	
213711	•	•		•			•		•				•
212707	•	•				•		•			•		
212709	•	•		•			•			•			•
219901	•					•		•		•		•	
219903	•		•		•	•		•	•				•
219904	•		•		•	•		•	•				•

FIRST YEAR

Course Code	Course Title	Marks	Credits
212801	Physical Chemistry-I	100	4
212803	Fundamentals of Organic Chemistry	100	4
212805	Fundamentals of Inorganic Chemistry	100	4
212806	**Part- A** : Chemistry Practical - I: Qualitative inorganic analysis	50	2
	Part- B : Chemistry Practical - II: Inorganic purification & preparations salts Elementary crystal chemistry	50	2
213709	Foundations of Mathematics	100	4
213711	Calculus-I	50	2
212707	Physics-I (Mechanics, Properties of Matter, Waves and Optics)	100	4
212709	Physics-II (Heat, Thermodynamics and Radiation)	50	2
219901	History of Bangladesh: Language, Culture and Identity	100	4
219903	Information and Communication Technology	**75**	3
219904	Lab: Information and Communication Technology	25	1
	Total =	**900**	**36**

Detailed Syllabus

Course Code	212801	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Course Title	Physical Chemistry-I			

Course Objective:

- The course will cover and as a support to understand the fundamental physical chemistry-I topics.
- To develop and extend student's knowledge on state of matter, properties of molecules, Liquid and Solution, chemical equilibrium, pH, chemical energetics and thermodynamics.

Course Learning Outcomes (CLOs):

After completing this course, students will be able to:

	Upon completion of this course, the students will be able to:		Mapping with PLOs
Course Learning Outcomes (CLOs)	**CLO1**	explain the fundamental knowledge about the characteristics of matter at different states.	A1
	CLO2	explain and apply laws that describe the behavior of gases.	B2
	CLO3	analyze and manipulate different parameters of gas.	A2, C2
	CLO4	compare different intermolecular forces and different states and chemical changes. pH and buffer solution	A2, B2
	CLO5	explain the vapor pressure of liquids and their variation with temperature	A1
	CLO6	analyze different real-life phenomena connected to liquids and gases, thermochemistry,	B2, C1, D2

Course Contents:

	Topics	Teaching Learning Strategy	CLOs
1	**State of Matter in Chemistry:** Nature and classification of matter. Classical states of solid, liquid, gas and Liquid crystals, state of aggregation of matter. Changes of state of matter, boiling and melting points. Intramolecular and intermolecular force and their role for the determination of state of matter. Modern state as plasma. Avogadro's number with applications, Concept of mole: calculation and importance.	Lecture, Group Discussion, Assignment	1
2	**Properties of Molecules**: Microscopic and macroscopic system. physical properties and their classifications, molar refractivity and dipole moment,		

	measurement, molar volume, magnetic rotation, optical exaltation, optical activity and molecular structure, potential and kinetic energy of molecules, degrees of freedom of motion; transnational rotational and vibrational energy of systems, principle of equipartition of energy.	Lecture, Group Discussion, Assignment	1, 2, 3
3	**Gaseous State:** Vapour and gas, measurable properties of gases, determination of gas pressure, the gas laws: derivation, molecular explanation and practical applications, effusion, diffusion, ideal and real gases, ideal gas equation with applications, kinetic theory and its application to ideal gases, pressure of ideal gases, Boltzmann distribution of molecular energies, deviation from ideal behavior and Boyle's temperature, Van der Walls equation, critical constants; continuity of state, principle of corresponding states; Liquefaction of gases, collision number, mean free path, Dalton's law of partial pressure. Heat capacities of gas.	Lecture, Group Discussion, Assignment	3
4	**Liquids and Solutions:** Liquids and their characteristics, Surface tension, measurement of surface tension, cohesion and adhesion force, capillary action, viscosity, Poiseuille's equation, measurement viscosity, evaporation, measurement vapour pressure, dependence of vapour pressure on temperature variation, heat of vaporization and Trouton's law, boiling, distillation, Types of solution, solubility and solubility curve, units of concentration for solution, Raoult's law, ideal and non-ideal solutions. Henry's law. a brief treatment of colligative properties, osmosis.	Lecture, Group Discussion, Assignment	2
5	**Chemical Equilibrium:** Equilibrium in chemical reactions. Stoichiometry, extant of reaction, law of mass action, equilibrium law and equilibrium constants: Kp, Kc, and Kx, effects of temperature, pressure and concentration changes of equilibrium, principle and applications of Le Chatelier and Braun, reaction quotient, extent of reaction, thermodynamic equilibrium constant, concentration-time curves.	Lecture, Group Discussion, Assignment	3, 4, 6
6	**pH and Buffer:** Ostwald dilution law, degrees of ionization and ionization constant, K_a, K_b, solubility product (K_{sp}) with principle and applications, common ion effect, ionic product of water (K_W), pH with applications, buffer solution, buffer mechanism, Henderson-Hasselbach equation, acid-base titration curves.	Lecture, Group Discussion, Assignment	5
7	**Chemical Energetics:** Importance and limitation of chemical thermodynamics, system, surrounding and boundary, heat and work, thermodynamic properties, state function and exact differentials, internal energy, enthalpy, thermodynamic process, work of expansions, reversible, irreversible and adiabatic expansions, heat capacities at constant pressure and constant volume, Joule-Thompson's effect, inversion temperature, enthalpy changes in various	Lecture, Group Discussion, Assignment	3, 6

	processes. First law of thermodynamics: Justification, mathematical expression & applications.		
8	**Thermochemistry:** Thermochemical equation, Heat of reaction, various type of reaction enthalpy, Hess's law and its application, heat of solution, bond enthalpy, atomization enthalpy, born-haber cycle.	Lecture, Group Discussion, Assignment	3, 5, 6

List of Books:

Atkins, P. (2024). *Physical chemistry*. W. H. Freeman and Company New York. USA

Castellan, G. W. (2024). *Physical Chemistry*. Addison-Wesley Publishing Company.

Chang, R. (2008). *General Chemistry: The Essential Concepts*. McGraw-Hill.

Ebbing, D. (2009). *General Chemistry*. Publisher Brooks/Cole Cengage Learning.

Glasstone, S. (2023). *A text Book of Physical Chemistry*. D. Van Nostrand Company.

Haque & Nawab (Revised by Haque & Mollah). (2009). *Principles of Physical Chemistry*. Published by Brothers' Publication 3/5, Rafine Plaza Mirpur Road, Dhaka-1205. Bangladesh.

Course Code	**212803**	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Course Title	**Fundamentals of Organic Chemistry**			

Course Ojectives:

- The course is designed for the student to impart and improve fundamental knowledge and aspects related to organic chemistry.
- To develop and extend student's knowledge on fundamental principles of organic chemistry bothe aliphatic and aromatic hydrocarbons, synthesis and reactivity of important functional groups, prepare for further studies of advanced organic courses, synthesize simple organic compounds and understand basics of reaction mechanism.

Course Learning Outcomes (CLOs):

After completing this course, students will be able to:

	Upon completion of this course, the students will be able to:		**Mapping with PLOs**
	CLO1	explain the historical background, fundamental concept, and scope and aim of organic chemistry.	A1, B1

Course Learning Outcomes (CLOs)	**CLO2**	clarify the bonding, functionalization, and isomerism involves in organic compounds.	A2
	CLO3	about the properties and nature of organic compounds and their interpretation.	A1, B1, D2
	CLO4	narrate different methods of preparation and reaction of organic compounds.	A1, A3, B1
	CLO5	conceptualize the mechanisms of major reaction involves in organic chemistry.	A2, A3

Course Contents:

	Topics	**Teaching Learning Strategy**	**CLOs**
1	**Basic Organic Chemistry:** Organic compounds, homologous series, the causes of diversity of organic compounds, Classification of organic compounds based on their functional groups, IUPAC nomenclature of organic compounds. Formation of carbocations, carbanion, free radicals and their stabilities. Applications of organic compounds.	Lecture, Group Discussion, Assignment	1, 2, 3
2	**Aliphatic Hydrocarbons:** **i. Alkanes:** Structure; nomenclature, sources; preparation and reaction of alkanes, conformation of n-butane, Wurtz reaction, free radical mechanism of halogenation. CFC (chloro-fluro carbon). **ii. Cycloalkanes:** Structure, nomenclature, sources preparation and reactions of cycloalkanes, small and normal size rings, Baeyer's strain theory. **iii. Alkenes:** Structure and orbital picture and sources, nomenclature, preparation and reaction of alkenes, geometrical isomerism of alkenes, cis-trans isomer, mechanism of electrophilic addition, Markowski's and Anti Markovnikov's rule. **iv. Dienes:** Structure and orbital picture, nomenclature; preparation and reactions of dienes, Diels-Alder reaction; polymerization, conjugated dines. **v. Alkynes:** Structure and orbital picture; nomenclature, preparation, reactions of alkynes, electrophilic addition reaction, acidity of alkynes.	Lecture, Group Discussion, Assignment	3, 4, 5

3	**Aromatic Hydrocarbons:** Source; structure, preparation and bonding in benzene; aromaticity; Huckell's rule, orientation in benzene ring and its derivatives, activation and deactivation in aromatic di-substitution, mechanism of electrophilic substitution in aromatic rings: halogenation, nitration, sulfonation, Friedel-Crafts alkylation and acylation	Lecture, Group Discussion, Assignment	3, 4, 5
4	**Alkyl and Aryl Halides:** Structure, nomenclature; preparation and properties of alkyl, aryl, allyl and vinyl halides; mechanism of nucleophilic substitution reaction and elimination reactions, Grignard reagent, its synthesis and application.	Lecture, Group Discussion, Assignment	2, 3, 4
5	**Alcohols and Phenols:** Structure, nomenclature, classification of alcohols, preparation, reactions of monohydric, dihydric and trihydric alcohols, reactions of alcohols, dehydration, substitution, oxidation and reduction, acidity and basicity of alcohols. Structure, synthesis and reactions of phenol, Acidity and reactivity of phenol and its derivatives.	Lecture, Group Discussion, Assignment	2, 3, 4, 5
6	**Ethers and Epoxides:** Nomenclature, preparation, reactions of aliphatic and aromatic ethers; ethers as protecting groups; crown ethers.	Lecture, Group Discussion, Assignment	2, 3, 4

List of Books:

Bahl, B. S., & Bahl, A. (2024). *Text- Book of Organic Chemistry*. S. Chand & Company Ltd Ram Nagar, New Delhi-1I005S. India.

English, J., & H.G Cassidy and Baird, R. I. (1949). *Principles of Organic Chemistry: An Introductory Text in Organic Chemistry*. MeGraw Hill.

Finar, I. L. (1973). *Organic Chemistry, Vol. I*. Pearson Education India.

Griffin, R. W. (2024). *Modern Organic Chemistry*. McGraw Hill.

Morrison, R.T. and Boyd, R. N. (1992). *Organic Chemistry*. Benjamin-Cummings Pub Co

Muny, J. Mc. (2010). *Organic Chemistry*. Thomson Broks-Coole.

Robert, J. D., & Caserio, M. C. (1977). *Basic Principles of Organic Chemistry*. W. A. Benjamin, Inc., Menlo Park, CA.

Solomons, T. W. G. (2008). *Fundamental of Organic Chemistry*. Wiley.

Course Code	212805	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Course Title	Fundamentals of Inorganic Chemistry			

Course Ojectives:

The objectives of this course are to:

- The course is designed for the student to impart and improve fundamental knowledge and aspects related to inorganic chemistry.
- To develop and extend student's knowledge on atomic structure, concept of periodic table, chemical bonding, acid and base, type of inorganic reaction, inorganic nomenclature and promote knowledge on the basic concepts of the inorganic chemistry to enter into the field of advanced inorganic chemistry.

Course Learning Outcomes (CLOs):

After completing this course, students will be able to:

	Upon completion of this course, the students will be able to:		Mapping with PLOs
Course Learning Outcomes (CLOs)	**CLO1**	discuss about the nature and composition of matter and various atomic theory.	A1, C3
	CLO2	explain electronic configuration of atoms with necessary principles and rules.	B1, C1
	CLO3	illustrate the classifications of the elements and their properties in the periodic table, acid and base	A4
	CLO4	classify bonds in common compounds along with covalent, ionic, and metallic bond	C3
	CLO5	explain the various theory about the structure of the molecules, nomenclature.	A4

Course Contents:

	Topics	Teaching Learning Strategy	CLOs
1	**Structure of atoms:** Fundamental particles, isotopes, atomic mass, mass cathode rays, mass and charge of an electron, α-particle scattering, Thompson atom model, Rutherford atom model, Planck's quantum theory, Bohr's theory for hydrogen atom, electromagnetic	Lecture, Group Discussion, Assignment	1

	spectrum, absorption and emission spectra, emission spectrum of atomic hydrogen, quantum numbers, atomic orbitals, shapes and orientation of s, p and d orbitals, Aufbau principle, Pauli exclusion principle.		
2	**Periodic Classification:** Periodic law, IUPAC periodic table, history of periodic table, prediction of elements, naming of all elements, elements in groups, periods and blocks, electronic configuration of groups and periods, metals, nonmetals and metalloids, diagonal relationship, periodicity of atomic and molecular properties e.g. ionization energy, electron affinity, electronegativity, effective nuclear charge, atomic/ionic radii, etc., usefulness and limitation of periodic table.	Lecture, Group Discussion, Assignment	1, 2
3	**Chemical Bonds:** Chemical bonds, types of chemical bonds, ionic bond: energetic of ionic bond formation, properties of ionic compounds, factors influencing the formation of ionic bond, Fajan's rule, radius ratio and crystal structure, covalent bond: single, double, triple, sigma and pi bonds, polar and non-polar covalent bonds, properties associated with covalent compounds, Lewis's formulation, formal charge, valence shell electron pair repulsion (VSEPR) theory and molecular geometry.	Lecture, Group Discussion, Assignment	1, 5
4	**Acids, Bases and Salts:** Theories of acids and bases, conjugate acids and bases, neutralization reactions, acid-base strength, leveling effect, hard and soft acids and bases, hard and soft acids and bases in qualitative analysis. Salts: Classification and their applications. Hydrolysis of salts: causes, degrees of hydrolysis, hydrolysis constant.	Lecture, Group Discussion, Assignment	3
5	**Chemical Reactions and their Classification:** Physical change and Chemical change, Methods of bringing about chemical reaction, Oxidation-reduction reactions, oxidizing and reducing agents, assigning oxidation states to bonded atoms, redox half reactions, rules for balancing redox reactions, standard reduction potential, the electrochemical series, disproportionation reactions, comproportionating reaction, addition reaction, elimination reactions, double decomposition reactions, substitution reactions, acid - base reactions.	Lecture, Group Discussion, Assignment	4, 5
6	**Inorganic Nomenclature:** Prefixes and affixes used in inorganic nomenclature, use of enclosing marks, numbers, letters and italic letters, names for cations, anions, radicals and heteropoly anions, names of acids, salts and salt like compounds.	Lecture, Group Discussion, Assignment	2, 4, 5

List of Books:

Cotton, F. A., Wilkinson, G., & Gaus, P. L. (1995). *Basic Inorganic Chemistry*. John Willey & Sons.

Ebbing, D. D. (2021). *General Chemistry*. Houghton Mifflin Co.

Madan, R. D. (2024). *Modern Inorganic Chemistry*. S. Chand & Company Ltd. New- Delhi. India.

Russel, J. B. (2024). *General Chemistry*. McGraw-Hill Inc.

Silberberg, M., & Amateis, P. (2017). *Chemistry – The Molecular Nature of Matter and Change*. McGraw-Hill Education.

Wulfsberg, G. (1991). *Principles of Descriptive Inorganic Chemistry*. University Science Books, Mill Valley.

Course Code	**212806**	**Marks: 50** **1 day (6 hours)**	**Credits: 2**	**Class Hours: 30 hrs**
Course Title	**Part- A** : **Chemistry Practical-I: Qualitative inorganic analysis**			

Course Ojectives:

To gain a fundamental knowledge on laboratory safety policy, some inorganic salts and radical analysis and some solution density measurement.

Course Learning Outcomes (CLOs):

After completing this course, students will be able to:

	Upon completion of this course, the students will be able to:		**Mapping with PLOs**
Course Learning Outcomes (CLOs)	**CLO1**	discuss about the general laboratory safety policy, rules and regulations and chemical management process.	A1, B1, D1, D2
	CLO2	explain the applications of apparatus and reagents in various experiments, different laboratory glassware and chemicals.	B1, D1, D2
	CLO3	prepare the primary and secondary standard substances and solution.	A4, B1, C3, D1, D2
	CLO4	know Qualitative analysis of mixtures of inorganic salts consisting of up to five different radicals.	A4, B1, C3, D1, D2

	CLO5	explain the nature and properties of inorganic insoluble substances, density mesurments.	A4, D1, D2

Course Contents:

	Topics	Teaching Learning Strategy	CLOs
1	**Safety:** Introduction of chemical, equipment's and safety in the laboratory.	Lecture, Group Discussion, Assignment	1, 2
2	**Introduction unit operations of chemistry laboratory:** Classifying chemicals, rules for handling reagents and solutions, cleaning and marking laboratory wares, analytical balance, sources of error in weighing, general rules of weighing, weighing types, weighing bottles, desiccators & desiccants, manipulating weighing bottles, weighing of solids & liquids, filtration and ignition of solids. Simple and filtering crucibles, filter paper, heating equipment's, rules for manipulating heated objects, maintenance & format of laboratory note book, laboratory materials & reagents, safety in laboratory, volumetric glass wares, calibration of volumetric glass wares, techniques, tolerance & precision of glass wares, centrifuge and filters, laboratory techniques for drying and dissolving, good laboratory practice, validation of analytical method, hierarchy of methodology quality assurance of analytical measurements handling of hazardous chemicals.	Lecture, Group Discussion, Experimental, Assignment	1, 2, 3
3	**Systematic Semimicro Qualitative Analysis of Inorganic Salts:** Identification of inorganic basic and acid (including interfering group) radicals, 5 radicals in a mixture taking at least two from both ion types.	Lecture, Group Discussion, Experimental, Assignment	4
4	**Density measurement:** Liquid, irregular solid by liquid.	Lecture, Group Discussion, Experimental, Assignment	5

List of Books:

Alexeyev, V. (2024). *Qualitative Analysis*. Mir Publishers.

Braun, D. R. (2024). *Introduction to Chemical Analysis*. McGraw Hill International,

Christian, G. D. (2023). *Analytical chemistry*. John Wiley & Sons.

Fifield, F. W., & Kealey, D. (2000). *Principles & practice of analytical chemistry*. Wiley, ISBN: 0632053844, 9780632053841

Skoog, D. A., & West, M. D. (2022). *Fundamental of analytical chemistry*. Publisher, Cengage

Learning; ISBN, 1285607198, 9781285607191.

Vogel, A. I. (2017). *A Text-Book of Macro and Semimicro Qualitative Inorganic Analysis*. Longmans, Green and Co. Ltd.

Welcher, F. J., & Hahn, R. B. (2024). *Semimicro Qualitative Analysis*. D. Van Nostrand Co. Inc

Wulfsberg, G. (1987). *Principles of Descriptive Inorganic Chemistry*. University Science Books, Mill Valley.

Course Code	**212806**	**Marks: 50** **1day (6 hours)**	**Credits: 2**	**Class Hours: 30 hrs**
Course Title	**Part- B : Chemistry Practical-II: Inorganic purification & preparations salts Elementary crystal chemistry**			

Course Objectives:

To gain a fundamental knowledge preparation and purification of some inorganic salts and model making for simple crystal systems.

Course Learning Outcomes (CLOs):

After completing this course, students will be able to:

	Upon completion of this course, the students will be able to:		**Mapping with PLOs**
Course Learning Outcomes (CLOs)	**CLO1**	discuss about the general laboratory safety policy, rules and regulations and chemical management process.	A1, B1, D1, D2
	CLO2	explain the applications of apparatus and reagents in various experiments, different laboratory glassware and chemicals.	B1, D1, D2
	CLO3	prepare stander solution.	A4, B1, C3, D1, D2
	CLO4	know synthesize some inorganic salts.	A4, B1, C3, D1, D2
	CLO5	explain to test purity of different salts by using simple and standard methods and crystalise with model making.	A4, D1, D2

Course Contents:

	Topics	Teaching Learning Strategy	CLOs
1	**Purification Inorganic Salts:** Purification of commercial NaCl by recrystallization and salting out processes	Lecture, Group Discussion, Experimental, Assignment	1, 2, 3
2	**Preparation of Inorganic Compounds:** Preparation of ferrous sulphate $FeSO_4.7H_2O$, Mohr's salt [$FeSO_4.(NH_4)_2SO_4.6H_2O$], potash alum [$K_2SO_4.Al_2(SO_4)_3.24H_2O$], chrome alum [$K_2SO_4.Cr_2(SO_4)_3.24H_2O$], sodium carbonate Na_2CO_3.	Lecture, Group Discussion, Experimental, Assignment	4
3	**Model Making:** Seven crystal systems model. Simple Molecular models.	Lecture, Group Discussion, Experimental, Assignment	5

List of Books:

Alexeyev, V. (2024). *Qualitative Analysis*. Mir Publishers.

Braun, D. R. (2024). *Introduction to Chemical Analysis*. McGraw Hill International,

Christian, G. D. (2023). *Analytical chemistry*. John Wiley & Sons.

Fifield, F. W., & Kealey, D. (2000). *Principles & practice of analytical chemistry*. Wiley, ISBN: 0632053844, 9780632053841

Skoog, D. A., & West, M. D. (2022). *Fundamental of analytical chemistry*. Publisher, Cengage Learning; ISBN, 1285607198, 9781285607191.

Vogel, A. I. (2017). *A Text-Book of Macro and Semimicro Qualitative Inorganic Analysis*. Longmans, Green and Co. Ltd.

Welcher, F. J., & Hahn, R. B. (2024). *Semimicro Qualitative Analysis*. D. Van Nostrand Co. Inc

Wulfsberg, G. (1987). *Principles of Descriptive Inorganic Chemistry*. University Science Books, Mill Valley.

Course Code	213709	Marks: 100	Credits: 4	Class Hours: 60
Course Title:	Foundations of Mathematics			

Course Description:

Foundations of Mathematics provides the basic tools required for higher studies in mathematics and related fields. The course covers relations and functions, real and complex number systems, polynomial equations, algebraic and geometric series, two- and three-dimensional geometry, and vectors.

Course Objectives:

To give students a clear understanding of essential mathematical concepts and techniques, preparing them for advanced courses in mathematics and allied disciplines.

Course Learning Outcomes (CLOs):

After completing this course, students will be able to:

CLO1	Define and explain basic concepts of relations, functions, and their graphs.
CLO2	Demonstrate understanding of real and complex numbers and apply their properties in problem-solving.
CLO3	Solve polynomial and related equations using appropriate methods.
CLO4	Apply formulas of algebraic and geometric series to compute sums and related results.
CLO5	Use the principles of two- and three-dimensional geometry to solve basic geometrical problems.
CLO6	Explain the basic concepts of vector space and apply them to simple problems.
CLO7	Represent and manipulate vectors in two and three dimensions to solve problems in mathematics and other fields.

Course Contents:

Topics	Teaching Learning Strategy	CLOs
Relations and functions: Relations; Order relation; Equivalence relations; Functions; Images and inverse images of sets; Injective, surjective, and bijective functions; Inverse functions.	Whiteboard-based lecture Assignment Short presentations by students	CLO1

Real Number System: Field and order properties; Prime numbers; Natural numbers; Integers and rational numbers; Absolute value and its properties; Basic inequalities (including inequalities of means and powers); Inequalities of Cauchy, Chebyshev, Weierstrass.	Whiteboard-based lecture Assignment	CLO2
Complex Number System: Field of Complex numbers; De Moivre's theorem and its applications.	Whiteboard-based lecture Assignment	CLO2
Matrices and Determinants: Notion of matrix; Algebra of matrices; Different types of matrices; Invertible matrices; Determinant function; Properties of determinants; Minors, Cofactors, expansion, and evaluation of determinants. Elementary row and column operations and row-reduced echelon matrices, Invertible matrices, Diagonal, triangular, and Symmetric matrices.	Whiteboard-based lecture Assignment	CLO3
System of Linear Equations: Linear equations; System of linear equations (homogeneous and non-homogeneous) and their solutions using different methods. Gaussian elimination, Application of matrices and determinants for solving systems of linear equations, Applications of systems of equations in real-life problems.	Whiteboard-based lecture Assignment	CLO3
Summation of finite series: Arithmetic and geometric series; Method of difference; Successive differences; Summation of trigonometric series.	Whiteboard-based lecture Assignment	CLO4
Theory of Equations: Relations between roots and coefficients; Symmetric functions of roots; Sum of the powers of roots; Synthetic division; Descartes' rule of signs; Multiplicity of roots; Transformation of equation.	Whiteboard-based lecture Assignment	CLO3
Two-dimensional Geometry: Transformation of coordinates, Pair of straight lines (homogeneous second-degree equations, General second-degree equations representing pair of straight lines, angle between pair of straight lines, Bisectors of angle between pair of straight lines), General equations of second degree (reduction to standard forms, Identifications, Properties and tracing of conics).	Whiteboard-based lecture Assignment	CLO5
Three-dimensional Geometry: Three-dimensional coordinates, Distance, Direction cosines and direction ratios, Planes and straight lines.	Whiteboard-based lecture Assignment	CLO5

Vector Geometry: Vectors in plane and space; Algebra of vectors; Rectangular components; Scalar and Vector products; Coplanar vectors; Scalar triple product and vector triple product; Applications of vectors to geometry (vector equations of straight lines and planes, areas and volumes).	Whiteboard-based lecture Assignment	CLO6 CLO7
Vector Spaces: Euclidean *n*-space, Real vector spaces, Subspaces, Linear combination of vectors, Linear dependence of vectors, Basis and dimension, Linear transformations, Matrix representation of linear transformation, Kernel and image, Eigenvalues and Eigenvectors.	Whiteboard-based lecture Assignment	CLO6 CLO7

List of Books:

Anton, H., & Rorres, C. (2013). *Elementary linear algebra with applications* (11th ed.). Wiley.

Barnard, S., & Child, J. M. (1997). *Higher algebra* (6th ed.). Cambridge University Press.

Hall, H. S., & Knight, S. R. (1992). *Higher algebra* (7th ed.). Macmillan.

Hummel, J. A. (1994). *Vector geometry*. Prentice Hall.

Khosh Mohammad. (2010). *Analytic geometry and vector analysis*. Dhaka: Ideal Library. Bangladesh.

Lipschutz, S. (1997). *Schaum's outline of set theory and related topics* (Schaum's Outline Series). McGraw-Hill.

Rahman, M. A. (2015). *Basic algebra*. Dhaka: Nahar Book Depot and Publications. Bangladesh.

Rahman, M. A. (2016). *Linear algebra*. Dhaka: Nahar Book Depot and Publications. Bangladesh.

Spiegel, M. R. (1974). *Vector analysis* (Schaum's Outline Series). McGraw-Hill.

Course Code	**213711**	**Marks: 50**	**Credits: 2**	**Class Hours: 30**
Course Title:	**Calculus-I**			

Course Description:

Calculus is one of the most fundamental courses in Mathematics, which consists of two parts (Differential and Integral). The course is designed to develop the topics of Differential and Integral calculus. Understanding this course will enable everyone to learn other mathematical courses that require the fundamentals of differentiation and integration.

Course Objectives:

To develop the basic ideas of functions and their graphs.

Learning the basic properties of limit and continuity and analyzing them both mathematically and graphically.

To understand the ideas and applications in solving real-life-oriented problems of differentiation and integration.

Course Learning Outcomes (CLOs):

After completing this course, students will be able to:

CLO1	Identify and graph various types of functions, including polynomial, rational, exponential, logarithmic, trigonometric, and hyperbolic functions, and describe their key properties.
CLO2	Explain the concepts of limits and continuity, and apply relevant theorems to compute limits.
CLO3	Apply techniques of differentiation, including rules and theorems (e.g., Leibniz's rule), to solve problems related to rates of change and approximations in mathematical and real-life contexts.
CLO4	Analyze functions using derivative-based tools, such as the Mean Value Theorem, to determine maximum and minimum values, and concavity, to solve optimization and curve analysis problems.
CLO5	Apply techniques of integration (definite and indefinite) and related theorems to solve problems involving area, volume, arc length, and surface area.
CLO6	Apply approximation techniques using Taylor polynomials and series to estimate function values and analyze convergence.
CLO7	Evaluate series expansions and perform differentiation and integration of series to solve complex calculus problems.

Course Contents:

Topics	Teaching Learning Strategy	CLOs
Functions and Their Graphs: Polynomial and rational functions; logarithmic and exponential functions; trigonometric functions and their inverses; hyperbolic functions and their inverses; combinations of such functions.	Whiteboard-based lecture Assignment	CLO1
Limit and Continuity: Definitions and basic theorems on limit and continuity; Limit at infinity and infinite limits; Computation of limits.	Whiteboard-based lecture Assignment	CLO2
Differentiation: Tangent lines and rates of change; Definition of derivative; One-sided derivatives; Rules of	Whiteboard-based lecture Assignment	CLO3

differentiation; Successive differentiation; Leibnitz's theorem; Related rates; Linear approximations and differentials.	Tutorial	
Applications of Differentiation: Mean value theorem; Maximum and minimum values of functions; Concavity and points of inflection; Optimization problems.	Whiteboard-based lecture Assignment	CLO4
Integration: Antiderivatives and indefinite integrals; Techniques of integration; Definite integration using antiderivatives; Fundamental theorems of calculus; Basic properties of integration; Integration by reduction.	Whiteboard-based lecture Assignment	CLO5
Applications of Integration: Arc length; Plane areas; Surfaces of revolution; Volumes of solids of revolution; Volumes by cylindrical shells; Volumes by cross sections.	Whiteboard-based lecture Assignment	CLO5
Approximation and Series: Taylor polynomials and series; Convergence of series; Taylor's series; Taylor's theorem and remainders; Differentiation and integration of series.	Whiteboard-based lecture Assignment	CLO6 CLO7

List of Books:

Anton, H., Bivens, I. C., & Davis, S. (2016). *Calculus: Early transcendentals* (11th ed.). Wiley.

Das, B. C., & Mukherjee, B. N. (1938). *Integral calculus*. Kolkata: U. N. Dhur & Sons Pvt. Ltd.

Das, B. C., & Mukherjee, B. N. (1949). *Differential calculus*. Kolkata: U. N. Dhur & Sons Pvt. Ltd.

Matin, M. A., & Chakraboty, B. (1994). *Differential calculus*. Dhaka: Standard Publications.

Mohammad, K., Bhattacharjee, P. K., & Latif, M. A. (1968). *Differential calculus* (1st ed.). Chittagong: S. Tripaty.

Mohammad, K., & Bhattacharjee, P. K. (1987). *Integral calculus* (6th ed.). Chittagong: H. Bhattacharjee.

Stewart, J. (2015). *Calculus: Early transcendentals* (8th ed.). Cengage Learning.

Swokowski, E. W. (1988). *Calculus with analytic geometry* (6th ed.). Brooks/Cole.

Thomas, G. B., & Finney, R. L. (1996). *Calculus and analytic geometry* (9th ed.). Addison-Wesley.

Course Code	212707	Marks: 100	Credits: 4	Class Hours: 60
Course Title	Physics-I (Mechanics, Properties of Matter, Waves and Optics)			

Course Objectives:

This course reviews the concepts of mechanics learnt at colleges from a more advanced perspective and goes on to build new concepts. It begins with Newton's Laws of Motion and ends with the Gravitation and Optics. Students will be taught the concept of Work, Energy, Power, Elasticity, Rotational Motion and Surface Tension. The students will be able to apply the concepts learnt to several real-world problems.

Course Learning Outcomes (CLO):

After going through the course, the student should be able to :

CLO1	Understand concept of Vector quantities and units
CLO2	Calculate work, energy and power
CLO3	Understand rotational motion
CLO4	Understand analyze gravitation
CLO5	Understand the laws of fluid dynamics
CLO6	Grasp the knowledge of physical properties of matter
CLO7	Understand the concept of waves and optics

Course Contents:

	Topic	Teaching Learning Strategy	CLOs
1	**Vector Analysis:** Vectors and scalars, Addition and multiplication of vectors, Triple scalar & vector products, Derivatives of vectors, Gradient, divergence and curl-their physical significance, Theorems of Gauss, Green & Stoke's.	Lecture, and Assignment	CLO1
2	**Work Energy and Power:** Work energy theorem, Conservation of energy and linear momentum, Conservative and non-conservative forces and systems, Conservation of energy and momentum, Centre of mass, Collision problems.	Lecture, Group Discussion, and Assignment	CLO2
3	**Rotational Motions:** Rotational variables, Rotation with constant angular acceleration, Relation between linear and angular kinematics, Torque on a particle, Angular momentum of a particle, kinetic energy of rotation and moment of inertia. Combined translational and rotational motion of a rigid body, Conservation of angular momentum.	Lecture, Group Discussion, and Assignment	CLO3

4	**Gravitation:** Centre of gravity of extended bodies, Gravitational field and potential their calculations, Determination of gravitation constant and gravity, Compound and Kater's pendulums, Motion of planets and satellites, Escape velocity.	Lecture, Group Discussion, and Assignment	CLO4
5	**Fluid Dynamics:** Viscosity and coefficient of viscosity, Poiseulle's equation, Determination of the coefficient of viscosity of liquid by Stoke's method, Bernoulli's theorem and its applications, Toricelli 's theorem, Venturimete.	Lecture, Group Discussion, and Assignment	CLO5
6	**Elasticity:** Moduli of elasticity, Poission's ratio, Relations between elastic constants and their determination, Cantilever.	Lecture, and Assignment	CLO6
7	**Surface Tension:** Surface tension as a molecular phenomenon, Surface tension and surface energy, Capillary rise or fall of liquids, Pressure on a curved membrane due to surface tension, Determination of surface tension of water, mercury and soap solution, Effect of temperature.	Lecture, and Assignment	CLO6
8	**Waves:** Mechanical waves, types of waves, travelling waves. The superposition principle. Wave speed, Power and intensity in wave motion. Interference of waves, Standing Waves and resonance.	Lecture, Group Discussion, and Assignment	CLO7
9	**Oscillatory Motions:** Simple harmonic motion, Combination of harmonic motions, damped harmonic motion, Forced oscillations and resonance.	Lecture and Assignment	CLO7
10	**Optics:** Fresnel's and Fraunhofer types, Diffraction through single slit and double slit, diffraction grating, Dispersive and resolving powers of gratings. Plane, Elliptical and circular Polarizations, Optical, Rotatory.	Lecture, Group Discussion, and Assignment	CLO7

List of Books:

Khan A. H., Ishaq. M. (2017). B.Sc. Hons. *Physics 1st Part (Bengali Version).* Ideal Books. Bangladesh.

Halliday, D., Resnick, R., & Walker, J. (2001). *Fundamentals of Physics.* 6th Edition John

Willey and Sons, Inc.

Sears & Zemansky's. (2015). *University Physics*: Vol-1, Mechanics, 12th Edition, Pearson.

Spiegel, M.R. (1959). Vector *Analysis*. McGraw-Hill.

Course Code	212709	Marks: 50	Credits: 2	Class Hours: 30
Course Title	Physics-II (Heat, Thermodynamics and Radiation)			

Course Ojectives:

The main objectives of the course are to teach the students the concepts of thermal energy, laws of thermodynamics and radiation.

Course Learning Outcomes (CLO):

After completing the course student will be able to:

CLO1: Understand basic concept of heat, temperature, and thermal equilibrium

CLO2: Calculate thermal energy by applying knowledge of calorimetry

CLO3: Comprehend the basic concepts of thermodynamics, the first and the second law of thermodynamics and 2nd law of thermodynamics.

CLO4: the concept of entropy and the associated theorems, the thermodynamic potentials and their physical interpretations

CLO5: Understand the basic aspects of kinetic theory of gases, Maxwell-Boltzman distribution law.

CLO6: Describe the **concept** of black body radiation, Emissive and absorptive powers, Rayleigh-Jean's law and applications of radiations laws.

Course Contents:

	Topic	Teaching Learning Strategy	CLOs
1	**Thermometry:** Temperature, concepts of thermal equilibrium, measurement of low and high temperature: Gas thermometers, Resistance thermometer, Thermocouple, Pyrometry, International temperature scale.	Lecture, and Assignment	**CLO1**
2	**Calorimetry:** Specific heats of solids, liquids and gases by method of mixture with radiation corrections: Newton's Law of cooling, Variations of specific heats, Atomic and molecular heats.	Lecture and Assignment	**CLO2**
3	**Thermodynamic Systems:** Concept of internal energy: The first	Lecture	**CLO3**

	law of thermodynamics, Work and specific heats, Isothermal and adiabatic processes.	and Assignment	
4	**The second law of Thermodynamics:** Reversible and irreversible processes: Carnot cycle, Efficiency of reversible engines, Absolute thermodynamic temperature scale, Change of phase: Clausius and Clapeyron equation, Porous plug experiment.	Lecture, Group Discussion, and Assignment	**CLO3**
5	**Entropy:** Entropy of an ideal gas, Temperature-entropy diagram, Increase of entropy.	Lecture	**CLO4**
6	**Thermodynamic Functions:** The Maxwell's relations, Specific heat equations.	Lecture	**CLO5**
7	**Radiation:** Concept of black body radiation, Kirchhoff's law, Stefan–Boltzmann law, Wien's displacement Law, Rayleigh-Jean's law, Planck's Radiation law, Temperature of the sun.	Lecture and Assignment	**CLO6**

List of Books:

Halliday, D., Resnick, R., & Walker, J. (2001). *Fundamentals of Physics.* 6th Edition John Willey and Sons, Inc.

Hossain, T. (1975) *Text Book of Heat.* Variety Books.

Khan, A.H. Ishaq, Mohammad. (2017). *B.Sc. Hons Physics 3rd Part. (Bengali Version*). Ideal Books. Bangladesh.

Saha, M.N. and Srivastava, B.N. A (1958) *Treatise on Heat.* The Indian Press Private Ltd. India.

Sears, F.W. (1950*). An Introduction to Thermodynamics.* Addison-Wesley Press.

Zemansky, M.W. (1997). *Heat and Thermodynamics.* 7th Edition. The McGRAW-HILL Co. Inc.

Course Code:	**219901**	**Marks: 100**	**Credits: 4**
Course Title:	**History of Bangladesh; Language, Culture and Identity**		**Hours: 60**

Course Objective:

This course is designed to help undergraduate students from diverse academic backgrounds develop a comprehensive and nuanced understanding of the historical development of Bangladesh, with a particular focus on its language, culture, and identity. By critically examining a variety of historical events, socio-political movements, and cultural shifts from ancient times to the contemporary period, students will have the opportunity to trace the evolution of the Bengali nation. The course aims to foster informed citizenship through an exploration of the Liberation War, identity formation processes,

cultural heritage, the lives and contributions of key political figures, and the roles of ethnic minorities. Ultimately, the objective is to equip learners with the intellectual tools necessary to contextualize current national debates and to recognize the role of youth and globalization in shaping the future of Bangladesh.

Course Learning Outcomes:

At the end of the course, learners will be able to:

CLO 1: Recall key historical events, dates, movements, personalities, and cultural developments that contributed to the emergence of Bangladesh from the pre-colonial period to the present day.

CLO 2: Demonstrate an understanding of the socio-political, linguistic, and cultural influences that shaped Bengali identity and nationalism, emphasizing the Language Movement, Liberation War, and subsequent democratic uprisings.

CLO 3: Apply their historical and cultural understanding to interpret contemporary national issues and debates relating to identity, political reform, and cultural transformation in Bangladesh.

CLO 4: Analyze how historical events, political ideologies, and social changes are interconnected in shaping the current socio-political landscape of Bangladesh, with a particular emphasis on the roles of movements, minorities, and the youth.

Course Contents:

Topic	Teaching Learning Strategy	CLOs
1. Pre-colonial Era • Life and Culture of the People in Ancient Bengal • Bengal Under Muslim Rulers: Society, Culture, and Religion • Role of the Sufis in Preaching Islam and Impacts of • Sufism on the Bengali Society • Bengali Society and Culture in the Writings of Foreigners	Interactive lectures	CLO 1-2
2. Colonial Era (18th and 19th Century) • The Battle of Plassey (1757) and the Beginning of British Colonialism • Bengal Renaissance • Reforms in Hindu Society • New Forms in Bengali Literature and Culture • The Muslim Response to Western Education	Interactive lectures, reading, and assignments	CLO 1-2
3. Colonial Era (First Half of the 20th Century) • Partition of Bengal (1905) • Hindu-Muslim Disagreements • Formation of the All-India Muslim League (AIML) • Muslim Shahitya Samaj • Buddhir Mukti Andolan: The Urge for Rational Thinking in Bengali Muslim Society • Growth of Religion-based Identity • Politics of Hindutva and the Two-Nation Theory • Spread of Communalism in Society	Interactive lectures and group discussions	CLO 2-3

• The Partition of India and Bengal		
4. Post-Partition Era (1947-1971) • Language Movement • Political, Economic, and Cultural Aspects • Growth of Vernacular Nationalism • Cultural Activism	Interactive lectures and group discussion	CLO 2-3
5. Changing Bengali Identity • The Evolution of Bengali Identity in the Context of Language, Culture, and Religion • From Ancient Times to the Present • The Role of the Bengali Language in Shaping Identity • The Language Movement of 1952 and Its Long-term Impact on National • Consciousness • The Influence of Religion on Bengali Identity • The Impact of Socio-political Movements on Identity Formation • The Liberation War of 1971 • Contemporary Debates on Bengali Identity • The Role of Youth in Redefining Identity • The Influence of Globalization	Interactive lectures and thematic assignments	CLO 2-3
6. Liberation Movement of 1971 and Mass Uprising till 2024 • The Political and Economic Exploitation of East Pakistan by West Pakistan • Six (6) Points Movement, Uprising of 1969 • The 1970 General Elections • The Non-cooperation Movement and the Declaration of Independence • The Role of AK Fazlul Haque, Huseyn Shaheed Suhrawardy, Maulana Bhashani, Sheikh Mujibur Rahman and Ziaur Rahman • The Liberation War of 1971 • Genocide and Resistance • The Role of the Mukti Bahini • Post-independence Challenges • Nation-building • Political Instability • Mass Uprisings and Democratic Movements • The Anti-autocracy Movement of the 1980s • The 1990s Movement for Democracy and Afterwards • The 2024 Movements for Political and Social Reforms	Interactive lectures, group discussions, and thematic assignments	CLO 1 CLO 3-4

7. History of Other Ethnic Groups • The Indigenous Communities of Bangladesh • Historical Presence • Cultural Practices • Land Rights and Cultural Assimilation • Contributions and Sacrifices • Political Representation of Ethnic Minorities	Interactive lectures and group discussions	CLO 4
8. Cultural Heritage and Modern Transformations • The Evolution of Bengali Culture • From Ancient Traditions to Modern Expressions • The Role of Literature, Music, and Art in Shaping Bengali Identity • Contributions of Rabindranath Tagore, Kazi Nazrul Islam, and Other Cultural Icons • The Impact of Globalization on Bengali Culture • The Revival of Traditional Arts • The Role of Youth in Cultural Innovation Urbanization • Environmental Changes • The Commodification of Culture	Interactive lectures, documentary screening, and thematic assignments	CLO 1-4

List of Books:

Ahmed, M. (1979). *Bangladesh: The constitutional quest for autonomy*, Dhaka: University Press Limited.

Ahmed, R. (1981). *The Bengal Muslims 1871-1906: A quest for identity.* Oxford University Press.

Alavi, H. (1972). *The state in post-colonial societies: Pakistan and Bangladesh.* In K. Gough & 11. P. Sharma (Eds.), Imperialism and revolution in South Asia (pp. 145-178). New York, NY: Monthly Review Press.

Bernier, F. (2023). *Travels in the Mogul Empire: A.D. 1656-1668.* Chennai: Atlantic Publishers.

Bleie, T. (2005). *Tribal peoples, nationalism, and the human rights challenge.* The Adivasis of Bangladesh. University Press Limited.

Bosc, S. (2011). *Dead reckoning Memories of the 1971 Bangladesh war*. Hurst & Company.

Eaton, R. M. (1996). *The rise of Islam and the Bengal frontier. 1204-1760.* Berkeley: University of California Press.

Gilmour, D. (2019), *The British in India: Three centuries of ambition and experience*. London. Penguin.

Habib, L. (1982). *Cambridge economic history of India. Cambridge.* Cambridge University Press.

Halim, S., Amanullah, A. S. M., & Nasir, R. 1. (Eds.). (2024). *Society and sociology in Bangladesh*: *A South Asian perspective.* The University Press Limited.

Hashmi, T. (2021). *Fifty Years of Bangladesh, 1971-2021 Crises of Culture, Development, Governance and Identity.* Switzerland: Palgrave Macmillan.

Huq, M. E. (1975). *History of Sufism in Bengal. Dhaka.* Bangladesh Asiatic Society

Husain, 1. (2014). *Karl Marx on India.* New Delhi: Tulika Books.

Jahan, R. (1972). *Pakistan: Failure in national integration.* New York, NY: Columbia University Press.

Maniruzzaman, T. (1988*). The Bangladesh revolution and its aftermath (2nd ed.)*. Dhaka: University Press Limited. (Original work published 19801

Mascarenhas, A. (1986). *Bangladesh: A legacy of blood. London:* Hodder and Stoughton.

Rashid, H. O. (2015). *The foreshadowing of Bangladesh: Bengal Muslim League and Muslim politics* (3rd ed.). Dhaka: University Press Limited.

Roy, A. (1984). *The Islamic syncretistic tradition in Bengal. Princeton.* NJ: Princeton University Press.

Sen, A. (2006). *Identity and violence: The illusion of destiny.* W. W. Norton & Company.

Umar, B. (2022). *The Emergency of Bangladesh: A History of East Pakistan.* Dhaka: Bangla Gobeshona.

Course Code	**219903**	**Marks: 75**	**Credits: 3**
Course Title:	**Information and Communication Technology**		

Course Objectives:

The main objective of the course is to develop students' understanding and skills in using, managing, and applying technology to solve problems and enhance various aspects of life and work. Key areas of focus include understanding ICT systems, software, hardware, networks, and their applications.

Course Learning Outcomes: The student will be able to:

CLO 1	Explain foundational ICT concepts, including the information processing cycle.
CLO 2	Operate standard computer hardware and software systems effectively.
CLO 3	Use office productivity tools (Word, Excel, PowerPoint) for academic and professional tasks.
CLO 4	Apply safe internet practices and use internet tools for communication and information retrieval.
CLO 5	Understand and explain the concepts of Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML)
CLO 6	Understand the concept of free-lancing

Course Contents:

Unit	**Specific Objectives**	**Content**	**Teaching and Learning Approach**

Unit 1: Introduction to Information and Communications Technology (ICT) and Computer System	The student will be able to: • Explain the concept of ICT and its related terminologies. • Describe the information processing cycle. • Analyze the impact of ICT on educational, social and economic development. • Identify career opportunities in ICT education. • Differentiate among the classes of computers and the usages of them. • Identify the vital components of the Systems Unit.	Definition of ICT, basic concepts and terminologies. Data, Information Application of ICT, Advantages and Disadvantages of ICT, Impact of ICT Career opportunities in ICT education Information Processing Cycle. Classification of Computers The Vital Components of the Systems Unit.	Classroom Lectures
Unit 2: Computer Hardware and Software	The student will be able to: • Describe the categories of Computer Hardware. • Describe the commonly used Input and Output devices • Identify the main processing devices, storage devices and media. Identify the main communication devices. • Identify types of Software packages. • Distinguish between an Operating System and Application software. • Identify different types, examples and uses of Operating Systems and Application software • Distinguish between Open Source and	Categories of Computer Hardware: Input devices, Processing devices, Output devices, Storage devices Communication devices, Main Processing Devices: The Processor, Control Unit and Arithmetic and Logic Unit Software Packages, Operating Systems, Types and uses of Operating Systems, Types and uses Application Software: Educational software, Games software, Graphics software Browsers: Internet explorer, Google chrome, Mozilla Firefox, Opera, Internet explorer, Mozilla Firefox; Proprietary and Open-Source Software	Classroom lectures and Lab

	Proprietary Software.		
Unit 3: Introduction to Word Processing Application	• The student will be able to: • Identify Word Processing packages. • Create and save a document using the Word Processor. • Format a Word document using formatting tools. Demonstrate the ability to perform collaborative editing. • Insert tables in a Word Processing document. • Insert symbols and pictures in Word Processing documents. • Use layout techniques in document creation. • Inserting headers and footers. • Print documents using the various print options.	Word Processing Packages and Their Uses. Introduction of Microsoft Word Software. File creation and saving: New, Save, Open, Save As. Editing: Copy/Cut, Paste, Undo, Redo, Find, Replace, Clipboard, Sub-Headings And Paragraphs, Text Correction, Wrapping Text, Text Orientation **Formatting:** Font: Style, Size, Color, Bold, Underline, Italic, Superscript, Subscript, Shadow, Strikeout. Paragraph: Alignment, Bullet & Numbering, Indent, Line Spacing, Table Border **Insert:** Table, Picture, Shape, Smart Art, Chart, Header, Footer, Page Number, WordArt, Drop Cap, Symbols, Equations, Hyperlink, Comment. **Page Design:** watermark, page border **Layout:** Page setup, print	Classroom Lectures, Lab, and Hands-on Practice
Unit 4: Spreadsheet Application	• The student will be able to: • Identify Spreadsheet Packages. • Explain the importance of the Spreadsheet application in data management. • Explain related concepts and terminologies in the Spreadsheet. • Identify features in the Spreadsheet	Spreadsheet Packages, e.g. Excel. Importance of Spreadsheet application in Data Management, Related Concepts and Terminologies (e.g., cell(s), rows, columns, worksheet, workbook) Excel Tool Bars: Formatting Bar, Standard Bar, Formula Bar. Types of Data and Their Uses (e.g., number, date, text, currency), Creating and Saving a Workbook, Formula and Functions, Sorting,	Classroom Lectures, Lab and Hands-on Practice

	application window. • Create and save a Workbook. • Construct and insert simple formulae and functions. • Format the worksheet using formatting tools. • Printing a worksheet.	filtering, validation, creating pivot table. Formatting Worksheet Using Formatting Tools. Draw Graph/chart, Printing Worksheet.	
Unit 5: Presentation Application	• The student will be able to: • State the importance of the Presentation application. • Identify the commonly used features of a Presentation application while prepare a presentation. • Create and save presentations using a template, Add new slide(s). Edit text, Format text, Insert objects, images and pictures, Run slide show, Apply transition, animation effects to slides • Select the print option for printing. • Prepare a presentation on a selected topic and present it.	Presentation Applications Packages, Devices used for Presenting, Importance of Presentation Application, Principles for Designing Presentations, Terminologies in Presentation Application (eg: Slide transitions, Slide show, etc.), Commonly Used Features of Presentation Application Window: Toolbars, Different presentation view modes Prepare a Presentation: Slide creating, editing, theme, background, inserting elements (image, audio, video hyperlink, table, smart art, graph, etc.) and formatting slides, Slide Layout, slide Show, slide transition, animation Effects. Selection of Print Option: Entire presentation, Specific slides, Handouts, Notes pages, outline view of slides.	Classroom Lectures, Lab and Hands-on Practice
Unit 6: Privacy and Security	The student will be able to: • Understand the basics of digital security • Use some security tools. • Understanding digital ethics.	Introduction to Information Security, cybercrime, DoS and DDoS Attack, Key Management, Digital Signature and Certifications, privacy, Data Security, Vulnerability, Threat and Risk, Malware, Social Engineering, Hacking, Plagiarism, Fishing, Software Piracy, Worms and Viruses, Spam, Adware, Malware, Spyware, Antivirus Software	Classroom Lectures, Lab and Hands-on Practice

		Ethics in the digital world	
Unit 7: Using The Internet to Communicate and Accessing Information	• The student will be able to: • Explain basic concepts, requirements, and terminologies of the Internet • Apply the rules and regulations in the use of the internet. • Using email • Use the internet social network to communicate. • Use Uniform Resource Locators (URLs) to access Information. Use search engines to access information • Upload files to virtual drives and work on it.	Internet, Intranet, Extranet, IP Address, Masking, MAC Address, Internet Services, OSI Reference Model, TCP/IP protocol stack, IPv4, IPv6, subnet Masking, MAC Address, Internet Services, Network Configuration and Troubleshooting, Wi-Fi, Broadband, Email Usage. Rules and Regulations in the Use of the Internet: Spam-Unsolicited Emails, People's Privacy, Intellectual Property Rights, etc. E-mail: Creating an Email Account, Sending, Accessing Email Messages, Attaching Documents to Email Messages, Using the Internet: Social Networks to Communicate, Uniform Resource Locators (URLs) to Access Information, Using Search Engines, Downloading Information from the Internet. Transferring Information from the Internet to a Different Application Intellectual Property Rights, ICT Policy, Software Piracy etc. Using Cloud Space: Google Drive, Google Workspace, OneDrive, Dropbox, etc.	Classroom Lectures, Lab and Hands-on Practice
Unit 8: Emerging Technologies: Data analytics, Artificial intelligence, Machine learning	• Define Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML). • Understand how data is collected, processed, and used for decision-making. • Recognize the role of	**Introduction to Data Analytics:** Define Data, Types of Data, Define Data Analytics, Basic Steps in Data Analytics, Applications of Data Analytics, Common Tools of Data Analytics (Excel, Tableau, Power BI, Python, SQL, etc.). **Artificial Intelligence (AI):**	Classroom Lectures, Lab and Hands-on Practice

	AI and ML in everyday life. • Explain basic differences between AI and ML. • Discuss benefits and challenges of these technologies. • Recognize current trends and career opportunities in these fields.	Define AI, Objectives of AI, AI capabilities, AI applications (NLP, Image Recognition, Predictive Analytics, Recommender Systems, Robotics, etc.); **AI Tools**: OpenAI GPT, TensorFlow, PyTorch, Keras, Scikit-learn,etc.), AI in daily life (Voice assistants, facial recognition, GPS route suggestions, chatbots, etc.) . **Machine Learning (ML):** Define Machine Learning (ML), How It Works, Classification of ML, applications of ML, and common ML platforms (Teachable Machine, TensorFlow, PyTorch, Microsoft Azure Machine Learning, etc.). Relationship Between Data Analytics, AI, and ML Benefits & Challenges of Data Analytics, AI, and ML. Future Trends & Career Paths.	
Unit 9: Introduction to Freelancing	The student will be able to: • Define and explain the fundamental concept and significance of freelancing in the global context. • Identify and categorize various online marketplaces and professional fields. • Develop professional profiles, portfolios, and proposals suitable for freelancing platforms. • Demonstrate the use of online payment systems and financial transaction	Introduction and Basic Concepts, Fields of Freelancing, Essential Skills and Preparation, Introduction to Freelancing Marketplaces, Creating Profile and Portfolio, Earning and Transaction Process, Building a Freelancing Career, Challenges and Solutions.	Classroom Lectures, Practical Demos, Assignments, Discussions

	methods. • Analyze and evaluate strategies for career development and effectively apply techniques to overcome freelancing challenges.		

List of Books:

Norton, P. (2008). *Introduction to computers* (9th ed.). The McGraw-Hill.

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.

Lambert, J., & Frye, C. (2015). *Microsoft Office 2016 step by step*. Microsoft Press.

Hassan, N. A., & Hijazi, R. (2017). *Digital privacy and security using Windows: A practical guide*. Apress.

Maheshwari, A. (2024). *Data analytics made accessible*. Jay Cobb.

Russell, S. J., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Alpaydin, E. (2020). *Introduction to machine learning* (4th ed.). MIT Press.

Course Code	**219904**	**Marks: 25**	**Credit: 1**
Course Title:	**Lab: Information and Communication Technology**		

Course Objectives:

The main objective of the course is to teach the students' understanding and skills in using, and applying technology to solve problems and enhance various aspects of life and work. It includes assembling hardware, installing software, preparing PPT slides, and producing Word and Excel documents.

Course Learning Outcomes:

The student will be able to:

CLO 1	Learn assembling hardware
CLO 2	Prepare, edit and print Word documents and excel
CLO 3	Prepare Power Point presentation.
CLO 4	Access information from Internet, e-mail
CLO 5	Installation of anti-virus software
CLO 6	Data collection, Prediction, Pattern recognition, classification
CLO 7	Exploring markets, Proposal writing, Simulation strategies

List of Experiments

CLO Addressed	Unit	Experiments	Teaching Learning Approach
CLO1	1-4	• Assemble different hardware • Install different software • Operate the computer - Drive, folder and file management • Maintenance	Lab and Hands-on Practice
CLO 2	3	**Word** • Prepare a Word document on a specific topic (e.g, routine, question paper, CV, reports, applications) • Formatting the document (Alignment, table, border, watermark, etc.), e.g., newspaper article, academic report, or documentation used in daily life, book, poster • Print documents with different paper and printers	Lab and Hands-on Practice
CLO 2	4	• **Excel** • Prepare a grade sheet • Prepare a family expenditure • Prepare a business expenditure report • Prepare payroll management, with a report • Create graphs on the given data • Print Excel files	Lab and Hands-on Practice
CLO 3	5	**Power point** • Prepare an academic presentation on a specific topic. • Formatting the slides & using different tools. • Apply animation and transition • Print PPT files in different modes: Hand note, Slides shorter, outline	Lab and Hands-on Practice
CLO4	6	• Install antivirus software, e.g., Norton Antivirus, McAfee, Kaspersky, Avast.	Lab and Hands-on Practice
CLO5	7	1. Use of email 2. Access information from the internet, use a search engine. 3. Use of virtual drive for collaboration 4. Google Meet	Lab and Hands-on Practice
CLO6	8	• Data Collection and Visualization (Excel) • Simple Prediction (Excel) • Pattern recognition (Teachable Machine) • Classification (Teachable Machine)	Lab and Hands-on Practice
CLO7	9	**Free lancing** • Marketplace exploration and portfolio creation • Proposal writing • Payment simulation	Lab, Hands-on, Group Work, Discussion

Instructions for the Lab:

1. Lab sessions should be hands-on, with students engaging in tasks like assembling hardware, installing software, and using office tools.
2. Software like Microsoft Word, Excel, PowerPoint, and Teachable Machine will be used for document creation, data analysis, and basic machine learning.
3. Labs also include antivirus installation, email use, and online collaboration via Google Meet and Drive, emphasizing teamwork and problem-solving.

Instructions for the Lab exam:

1. The lab exam is individual. Students must answer two out of seven questions.
2. Marks distribution: Experiment (20 marks, 10 for each) for practical tasks, Viva-Voce (5 marks) for oral assessment. Total: 25 marks.

List of Books:

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.

Lambert, J., & Frye, C. (2015). *Microsoft office 2016 step by step*. Microsoft Press